IPL Player Performance & Auction Value Prediction

Team Owner Report

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Executive Summary

This project uses IPL player performance statistics to estimate auction value. The workflow combines SQL Server for data preparation with Python and machine learning for prediction. The final machine-learning dataset contains 39 players with valid auction or RTM purchase values. Virat Kohli was not used for model training because the available INR 0.12 crore value was an U19 draft value rather than a genuine auction/RTM purchase. The project therefore keeps the target definition consistent across the modeling data.

What the Model Uses

The model uses 12 performance features: matches played, total runs, balls faced, total fours, total sixes, strike rate, centuries, half-centuries, ducks, total wickets, bowling economy, and performance score. The target is auction value measured in crores. The idea is straightforward: give the model a player performance profile, and it estimates an auction value based on patterns learned from the historical examples.

Model Comparison

Model	MAE (Cr)	RMSE (Cr)	R2
Linear Regression	2.900	4.327	-0.217
Random Forest	2.849	4.167	-0.129
XGBoost	2.997	4.593	-0.371
Tuned Random Forest	**2.659**	**3.907**	**0.008**

The tuned Random Forest is the best model among the tested approaches because it has the lowest MAE and RMSE and the highest R2. Its settings were 200 trees, maximum depth 2, minimum samples per leaf 3, and random_state 42.

What Matters Most

The final tuned Random Forest ranked strike rate as the most influential feature, followed by matches played, total sixes, bowling economy, and performance score. For a team owner, this suggests that the model is paying particular attention to scoring speed, experience, power hitting, bowling efficiency, and the overall performance measure.

Business Interpretation

A team could use a system like this as a screening or discussion tool during player evaluation. Instead of looking at one statistic in isolation, the model considers several batting and bowling indicators together. For example, a player with strong strike rate and power-hitting numbers may receive a higher estimated value, while a player with weaker overall performance may receive a lower estimate. The model can also help compare candidates using a consistent numerical framework.

Important Limitation

The model should not be treated as a guaranteed auction-price predictor. The ML dataset contains 39 players, with 31 used for training and 8 for testing. Auction values are also influenced by factors that are not present in the current feature set, such as team requirements, player availability, recent form, age, role demand, competition during the auction, and market conditions. The R2 value of 0.008 confirms that the current model explains very little of the variation in the test-set auction values. The system is therefore best presented as an academic estimation model and a demonstration of the SQL-to-ML workflow.

Conclusion

The project demonstrates an end-to-end sports analytics workflow. SQL organizes the IPL data and creates player-level performance information, while Python and machine learning use that information to estimate auction value. The tuned Random Forest is the strongest of the tested models, but the current results also show that performance statistics alone are not enough for highly accurate real-world auction valuation. A larger dataset and additional contextual features would be the natural next improvements.